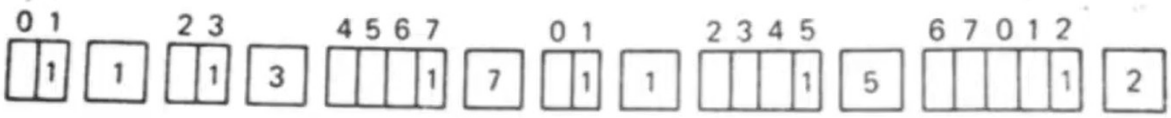
BRAP (BROADCAST RECOGNITION WITH ALTERNATING PRIORITIES)

- BRAP (Broadcast Recognition with Alternating Priorities) was discovered independently by Chlamtac (1976) and Scholl (1976).
- Chlamtac's version is called BRAM (Broadcast Recognition Access Method); Scholl's version is called MSAP (Mini Slotted Alternating Priorities). As a compromise, the name BRAP is adopted.
- BRAP (Broadcast Recognition with Alternating Priorities) is a collision free protocol that operates in the Medium Access Control (MAC) layer of the OSI model.
- It resolves any possibility of collisions while multiple stations are contending for acquiring a shared channel for transmission.
- In BRAP, a contention period containing variable slot rotates among the stations in a round-robin fashion.
- As soon as a station inserts a 1 bit into its slot, it begins transmission of its frame immediately thereafter.
- In addition, instead of starting the bit scan with station 0 each time, it is started with the station following the one that just transmitted.



- Each contention period always consists of a run of (zero or more) empty slots, terminated by a single 1 bit.
- As soon as someone announces that he wants the channel, he gets it immediately.
- BRAP can also be described in a slightly different form.
- Going back to 1-persistent CSMA, we could say that BRAP consists of having each station delay its attempt to seize the channel, with the delay time being proportional to the difference between the station's number and the number of the last successful transmission, modulo N.
- Since the delays are staggered, there are no collisions.
- Although the channel efficiency of BRAP is identical to the basic bit-map method, its delay characteristics are better.
- In particular, at low load a station only has to wait an average of (N/2)-bit slots before starting (for large N).
- At high load, BRAP and the basic method are similar, since the major delay comes from the frames, not the contention slots.